

# Software Requirements Specification (SRS) Version 1

AgrilabVision \_version 1 by BAKARY COULIBALY

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# 1. Introduction

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the functional and non-functional requirements of AgrilabVision, a mobile application I am developing to assist farmers in analyzing soil characteristics and determining agricultural recommendations.

This document explains what the system shall do, how it must behave, and the constraints under which it must operate. It serves as the official technical reference for myself, Coulibaly Bakary, and for any future collaborators, testers, instructors, or stakeholders.

## 1.2 Document Conventions

“I”, “me”, and “my” refer to myself, Coulibaly Bakary, the developer of this project.

Functional requirements follow the notation FR-XX.

Nonfunctional requirements follow the notation NFR-XX.

Priorities are expressed as:

- H = High priority

- M = Medium priority
- L = Low priority

## 1.3 Intended Audience and Reading Suggestions

This SRS is intended for:

Instructors, who will evaluate the completeness and clarity of the document.

Myself, as the project developer responsible for implementation.

Testers, who will validate the system against its stated requirements.

Future developers or collaborators, who may extend or modify the system.

Readers should first review Sections 1 and 2 for contextual understanding, then proceed to Section 3 for detailed requirements and Section 5 for system constraints.

## 1.4 Project Scope

AgrilabVision is a mobile application designed to help farmers in rural communities assess soil fertility using only a smartphone camera.

The system will:

capture soil images,

analyze soil texture (sand, clay, silt),

detect soil moisture level,

determine the agricultural season,

provide crop recommendations,

recommend appropriate fertilizers,

offer irrigation guidance,

work offline,

and store analysis history.

The main objective is to make agricultural decision-making more accessible for communities lacking laboratory resources or agricultural experts.

## 1.5 References

Official Flutter Documentation

TensorFlow Lite Documentation

FAO Soil Classification Guidelines

OpenWeatherMap API Documentation

AgrilabVision Project Charter (2025)

## 2. Overall Description

### 2.1 Product Perspective

AgrilabVision is an independent mobile application developed using the Flutter framework. It integrates several technologies:

a TensorFlow Lite machine-learning model for on-device soil image analysis,

a weather API to determine local agricultural seasons,

a local SQLite database for data persistence,

optional cloud synchronization through Firebase when connectivity is available.

This application does not depend on any other mobile software and is designed to run fully offline.

### 2.2 Product Features

AgrilabVision will include the following major functionalities:

Soil image capture through the smartphone camera

Soil texture analysis using an AI model

Soil moisture detection

Weather-based season detection

Crop recommendations

Fertilizer recommendations

Irrigation advice

User profile and farm plot management

Analysis history view

Full offline mode with eventual synchronization

## 2.3 User Classes and Characteristics

| User Class | Description |
|------------|-------------|
|------------|-------------|

|        |                                                                  |
|--------|------------------------------------------------------------------|
| Farmer | Has limited technical knowledge; needs simple and clear outputs. |
|--------|------------------------------------------------------------------|

|                      |                                                      |
|----------------------|------------------------------------------------------|
| Agricultural Officer | Manages analyses for multiple farmers or farm plots. |
|----------------------|------------------------------------------------------|

|                    |                                                      |
|--------------------|------------------------------------------------------|
| Student/Researcher | Uses the system for academic or scientific purposes. |
|--------------------|------------------------------------------------------|

## 2.4 Operating Environment

AgrilabVision will operate under the following conditions:

Android devices running Android 7.0 or later,

Minimum 8MP smartphone camera,

Flutter mobile framework,

TensorFlow Lite model running on-device,

Local storage via SQLite or SharedPreferences,

Optional cloud environment provided by Firebase.

## 2.5 Design and Implementation Constraints

The application must run effectively in areas with poor or no Internet connectivity.

AI model size must not exceed 50MB for performance reasons.

Smartphone camera quality may significantly influence soil analysis accuracy.

Local device storage limitations may impact history size.

The system must comply with Android permission policies for camera and storage.

## 2.6 User Documentation

The final system shall include:

a user tutorial presented during first use,

a help section explaining features,

an FAQ page,

visual onboarding steps.

## 2.7 Assumptions and Dependencies

Assumptions:

Users have basic farming knowledge.

Users have access to Android smartphones.

Dependencies:

Weather API availability and uptime,

Reliability of the TensorFlow Lite model,

Proper camera and storage permissions granted by the user.

## 3. System Features

### 3.1 Soil Image Capture

Description:

This feature allows users to capture soil images using the smartphone camera.

Functional Requirements

FR-01: The system shall allow the user to open the device camera from within the application.

FR-02: The system shall allow the user to capture an image of soil.

FR-03: The system shall store each captured soil image locally on the device.

## 3.2 Soil Texture Analysis

### Functional Requirements

FR-04: The system shall analyze the captured soil image to determine soil texture (sand, clay, silt).

FR-05: The system shall complete soil texture analysis in less than five seconds.

## 3.3 Moisture Detection

### Functional Requirements

FR-06: The system shall estimate the moisture level of the soil based on the captured image.

FR-07: The system shall classify the soil as dry, moist, or wet.

## 3.4 Season Detection

### Functional Requirements

FR-08: The system shall retrieve weather data through an external weather API.

FR-09: The system shall determine the agricultural season based on weather and regional indicators.

## 3.5 Crop Recommendation

### Functional Requirements

FR-10: The system shall generate crop recommendations based on soil texture, moisture, and season.

FR-11: The system shall provide at least three crop options suitable for the identified soil conditions.

## 3.6 Fertilizer Recommendation

### Functional Requirements

FR-12: The system shall identify potential nutrient deficiencies using soil analysis results.

FR-13: The system shall recommend fertilizers appropriate to the selected crop.

## 3.7 Irrigation Guidance

### Functional Requirements

FR-14: The system shall provide irrigation recommendations based on soil moisture level and weather conditions.

## 3.8 Offline Mode

### Functional Requirements

FR-15: The system shall operate without Internet access.

FR-16: The system shall automatically synchronize stored data when Internet connectivity is restored.

## 3.9 User Profile

### Functional Requirements

FR-17: The system shall allow users to create and manage personal profiles.

FR-18: The system shall allow users to register farm plots and save plot data.

## 3.10 Analysis History

### Functional Requirements

FR-19: The system shall store all soil analyses in a local history.

FR-20: The system shall display past analyses chronologically.

# 4. External Interface Requirements

## 4.1 User Interfaces

The user interface shall include:

Buttons at least 48×48dp, following accessibility guidelines.

Text elements sized 16sp or greater for readability.

Icons accompanied by text labels.

A dashboard displaying the main actions:

- Capture Image
- Analyze Soil
- View History
- Settings

## 4.2 Hardware Interfaces

Smartphone camera

Internal storage

GPS sensor (optional)

## 4.3 Software Interfaces

TensorFlow Lite inference engine

SQLite local database

Weather API (REST/JSON)

Firebase cloud sync service (optional)

## 4.4 Communication Interfaces

HTTPS protocol

REST API communication

JSON data format

## 5. Other Nonfunctional Requirements

### 5.1 Performance Requirements

NFR-01: Soil analysis must complete in under five seconds.

NFR-02: Weather API calls must return in under two seconds.

NFR-03: Application startup time must not exceed four seconds.

### 5.2 Safety Requirements

NFR-04: The system must avoid providing harmful agricultural advice.

NFR-05: The system shall notify the user when analysis confidence is low.

### 5.3 Security Requirements

NFR-06: All user data shall be encrypted locally and during synchronization.

NFR-07: Cloud synchronization shall require user authentication.

## 5.4 Software Quality Attributes

The system must be:

Usable, with intuitive navigation.

Reliable, with minimal crashes.

Maintainable, with modular code.

Efficient, operating well on low-end devices.

Portable, with potential for future iOS support.

Multilingual support (English and French).

Expandable database for crops and fertilizer types.

## Appendix A: Glossary

Soil Texture: Ratio of sand, clay, and silt.

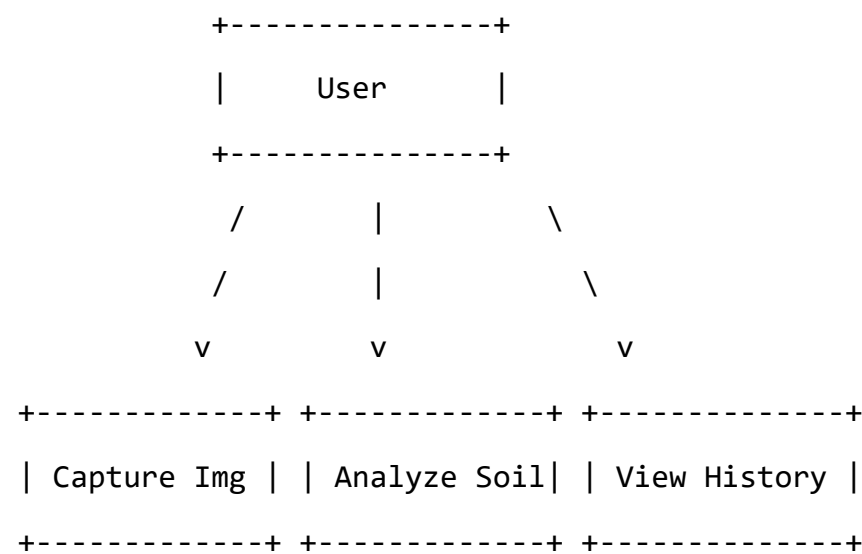
Moisture Level: Observable water content in soil.

TensorFlow Lite: Lightweight on-device machine-learning model.

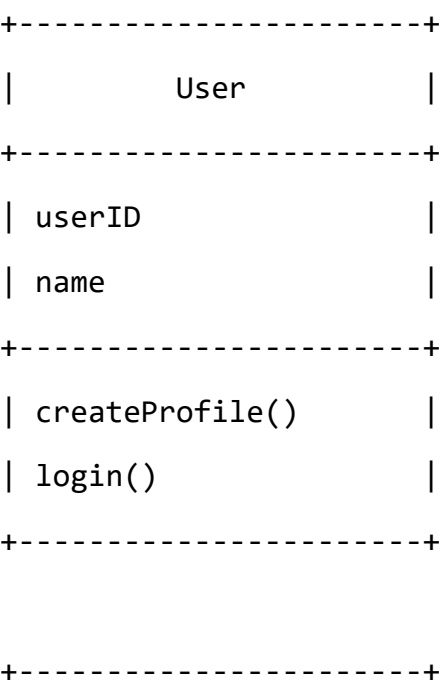
Offline Mode: Ability to function without Internet connection.

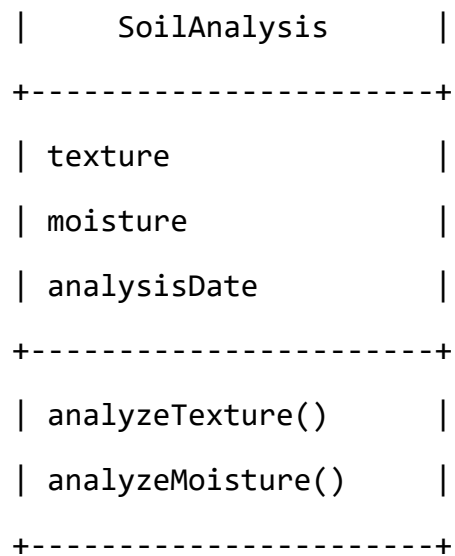
## Appendix B: Diagrams

Use Case Diagram (ASCII)

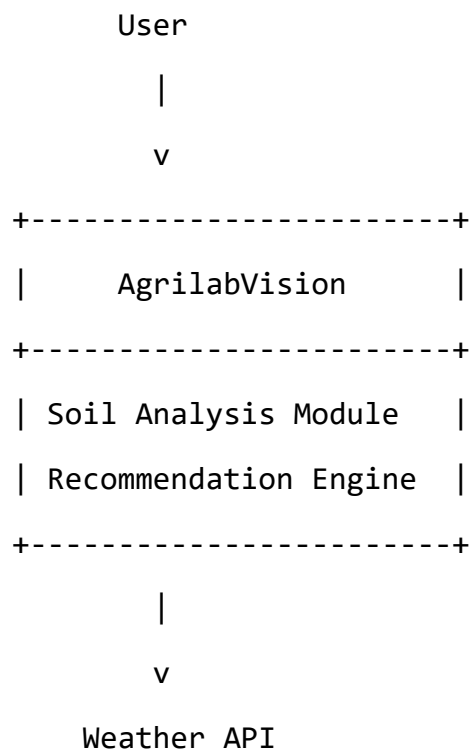


Class Diagram (ASCII)





DFD Level 0 (ASCII)



## Appendix C: Issues List

Soil dataset requires expansion.

Machine-learning model requires validation by agricultural experts.

Offline mode must be tested across low-end Android devices.